

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Disk

Lecture No.- 2

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Recap of Previous Lecture



Doubt

Topic

Magnetic Disk

Topic

Disk Capacity

Topic

Disk Access Time



Topics to be Covered



Topic

Multiple Sector Access

Topic

Disk Cylinder

Topic

Disk Addressing

[MSQ]

A, B, D

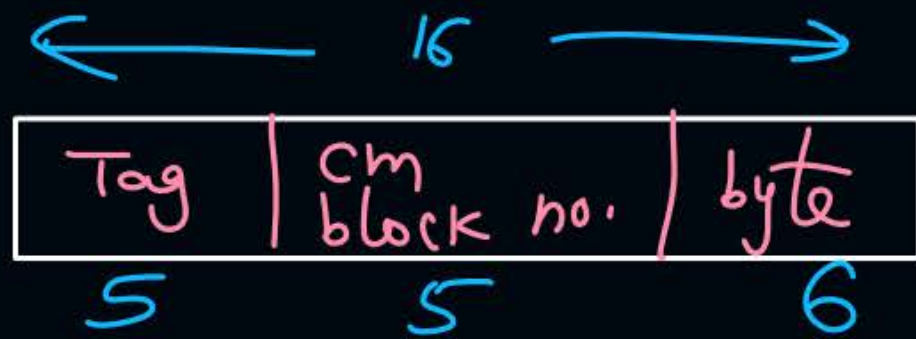
$2^{16} \text{ B} \Rightarrow \text{memory address} = 16 \text{ bits}$

[2022 : 2M]



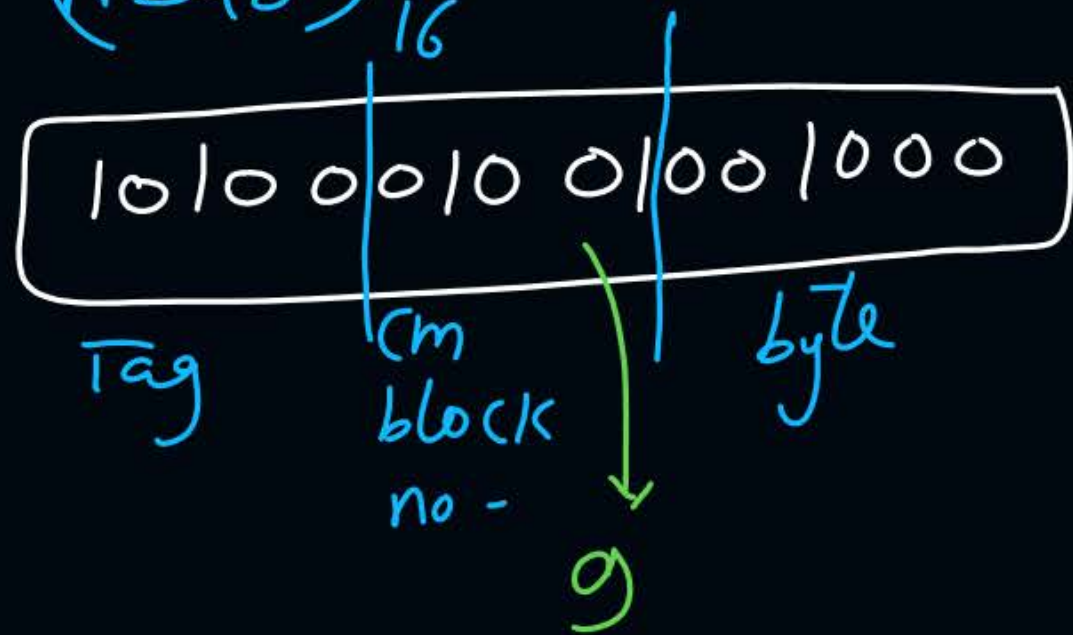
#Q. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

- ☒ **A** Every access to S is a hit.
- ☒ **B** Once P is brought to the cache it is never evicted.
- ☐ **C** At the end of the execution only R and S reside in the cache.
- ☒ **D** Every access to R evicts Q from the cache.

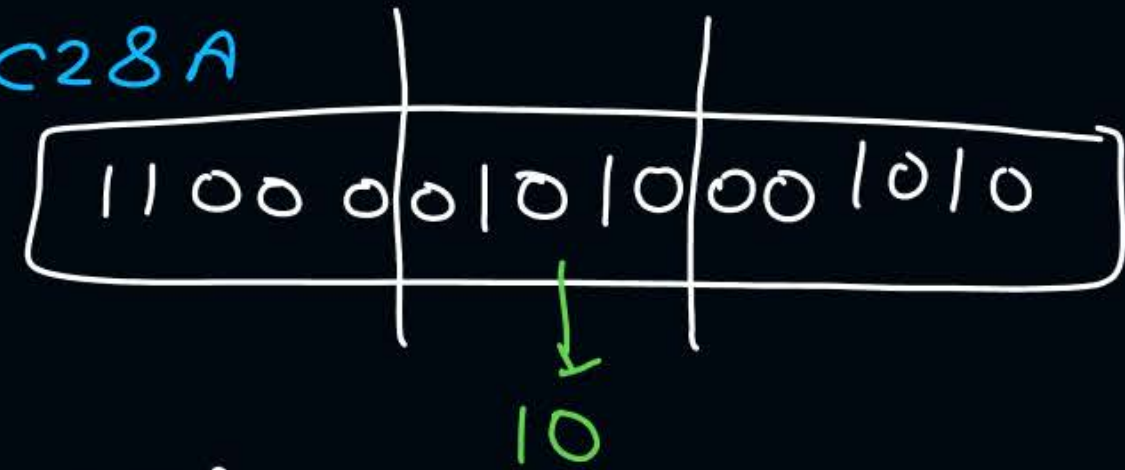


$\log_2 k = 11$

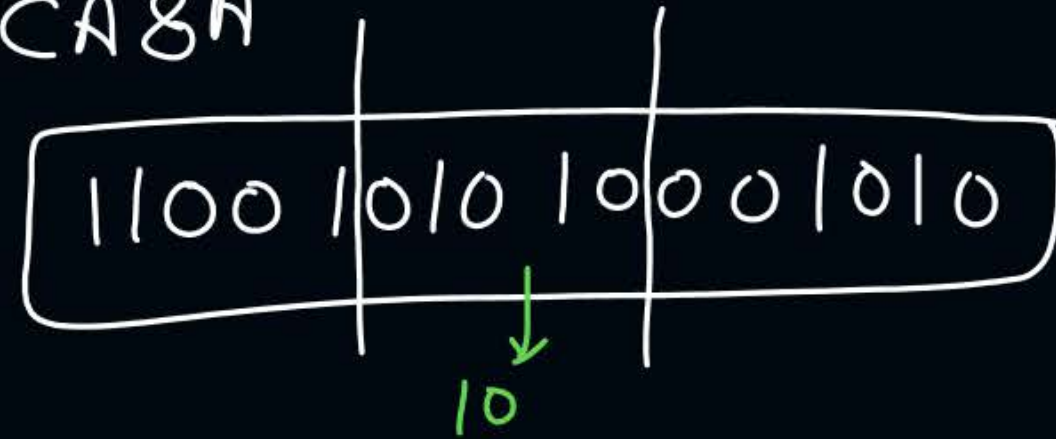
P:- (A248)



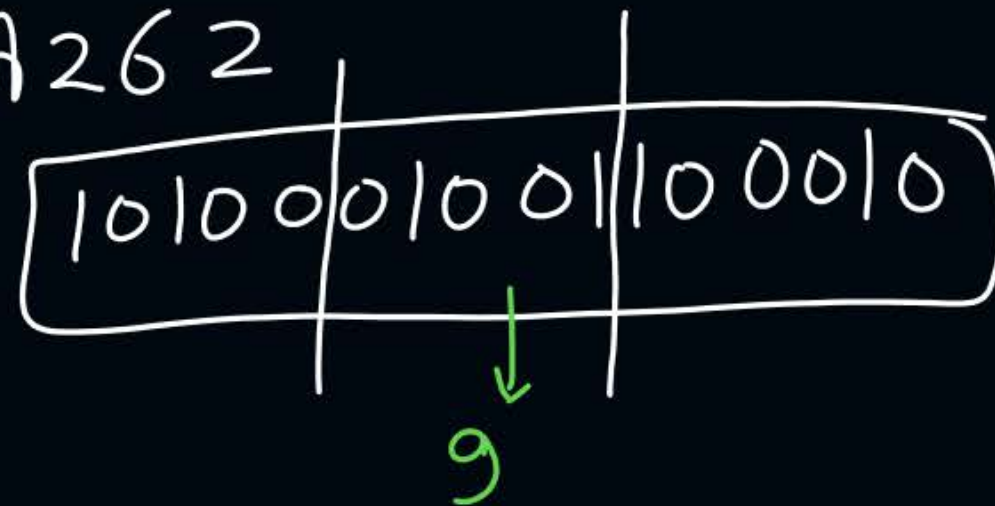
Q:- C28A



R:- CA8A



S:- A262



Ques) In prev. Questⁿ no. of hits and miss ?

	hits	miss
P	9	1
Q	0	10
R	0	10
S	10	0
	19	21

Ans = 40 %

#Q. Consider a disk with 16 platters, 2 surfaces per platter, 1K tracks per surface, 2K sectors per track and 2048 Bytes per sector. Disk rotates with 3000 rpm. Seek time is 10ms.

1 rotation
20 msec

If the disk is used in cycle stealing mode of DMA, such that whenever 64^{8B} bits word is available, it will be transferred in 16ns. What is the % of time CPU is blocked?

$$1 \text{ sector Transfer time} = \frac{20 \text{ ms}}{2 \text{ K}} = 0.01 \text{ msec}$$

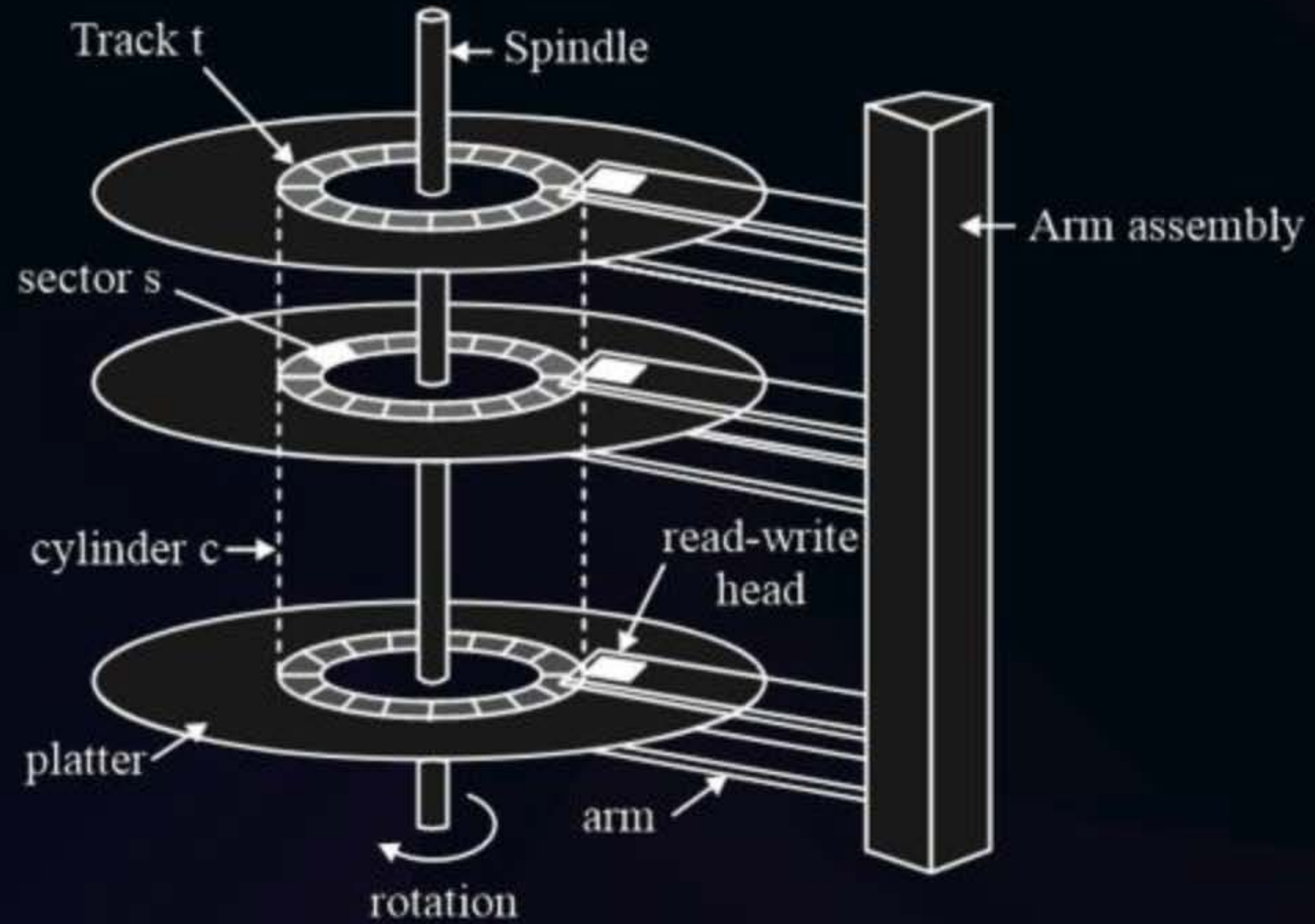
$$\text{for } 2 \text{ KB Preparation} = 0.01 \text{ ms}$$

$$\text{for } 8 \text{ B} \text{ --- } = \frac{0.01 \text{ ms}}{2 \text{ KB}} * 8 \text{ B} = 0.04 \text{ usec} = 40 \text{ ns}$$

$$\begin{aligned} \% \text{ of time CPU blocked} &= \frac{\overset{4}{\cancel{16}}}{\underset{\cancel{10}}{\cancel{40}}} * 100\% \\ &= 40\% \end{aligned}$$



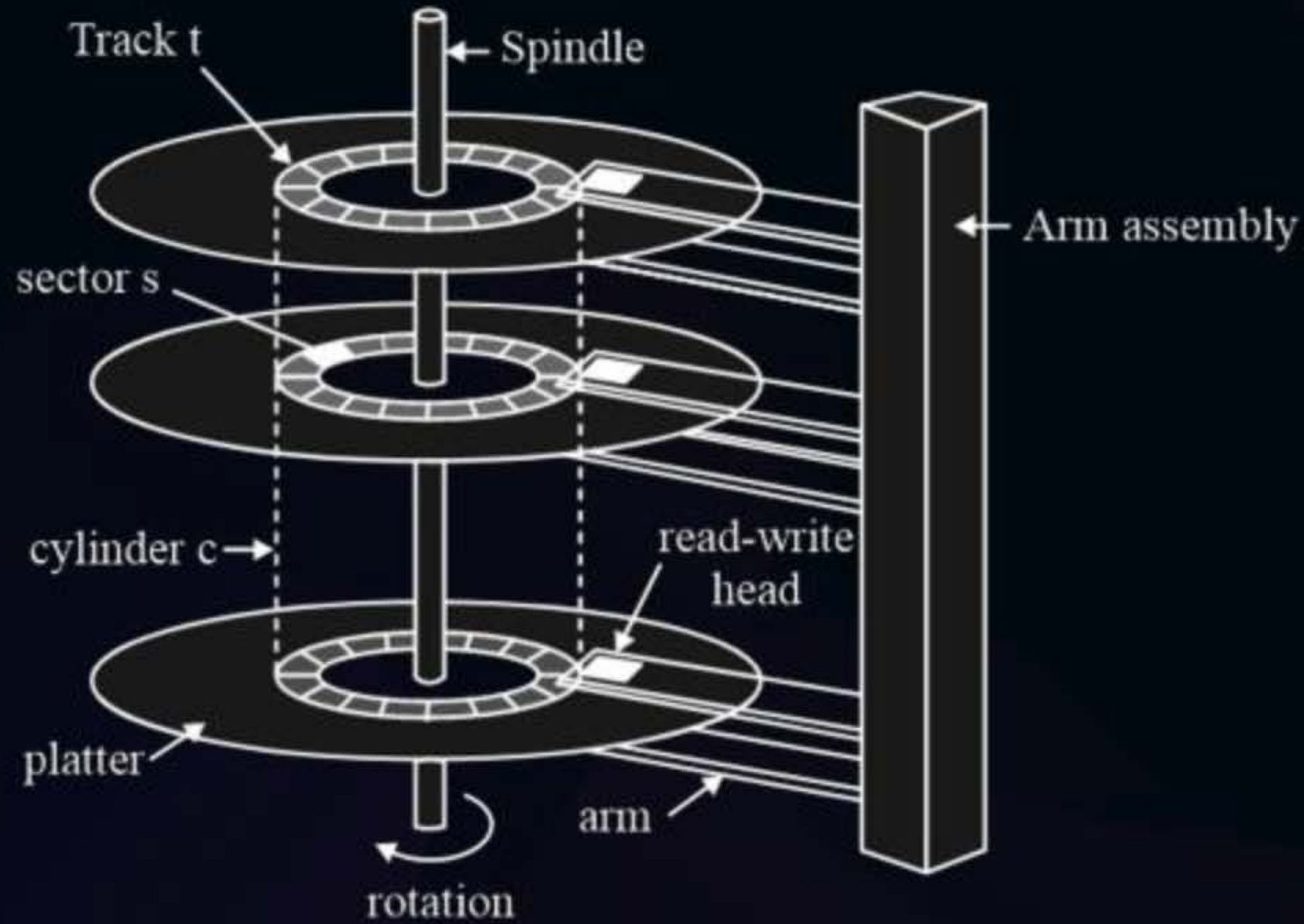
Topic : Multiple Sector Access Time





Topic : Multiple Sectors Access Time: Sequential

↓
on a track





Topic : Multiple Sectors Access Time: Sequential

Consider n sectors to be transferred:



$$= \text{seek time} + \text{rotational latency} + n * \text{1 sector transfer time}$$



Topic : Multiple Sectors Access Time: Random

Consider n sectors to be transferred:

$$= n * \left[\text{seek time} + \text{rotational latency} + \frac{1}{\text{sector transfer time}} \right]$$

1 rotation time = 6 ms

GATE-PYQ

#Q. Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is

A 14020

B 14000

C 25030

D 15000

$$1 \text{ rotation time} = 10 \text{ ms}$$

#Q. An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10ms. Rotational speed of disk is 6000rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected)

(A) 0.5 s

(B) 1.25 s

☒ (C) 1.5 s

(D) 1.0 s

$$= 100 * \left[10 + \frac{10}{2} + 0 \right] \text{ ms}$$

$$= 1500 \text{ ms}$$

$$= 1.5 \text{ sec}$$

Ques) If disk transfer rate is 2 MB/sec
then time needed to read a sector of $4\text{ KB} = ?$

Solⁿ

for 2 MB data time = 1 sec

$$\text{for } 4\text{ KB} \text{ ---||---} = \frac{1\text{ sec}}{\cancel{2\text{ MB}}_k} * \cancel{4\text{ KB}}^2$$
$$= 2\text{ msec}$$

#Q. Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is ____?





Topic : Cylinder



collection of all the same radius tracks from all surfaces forms a cylinder



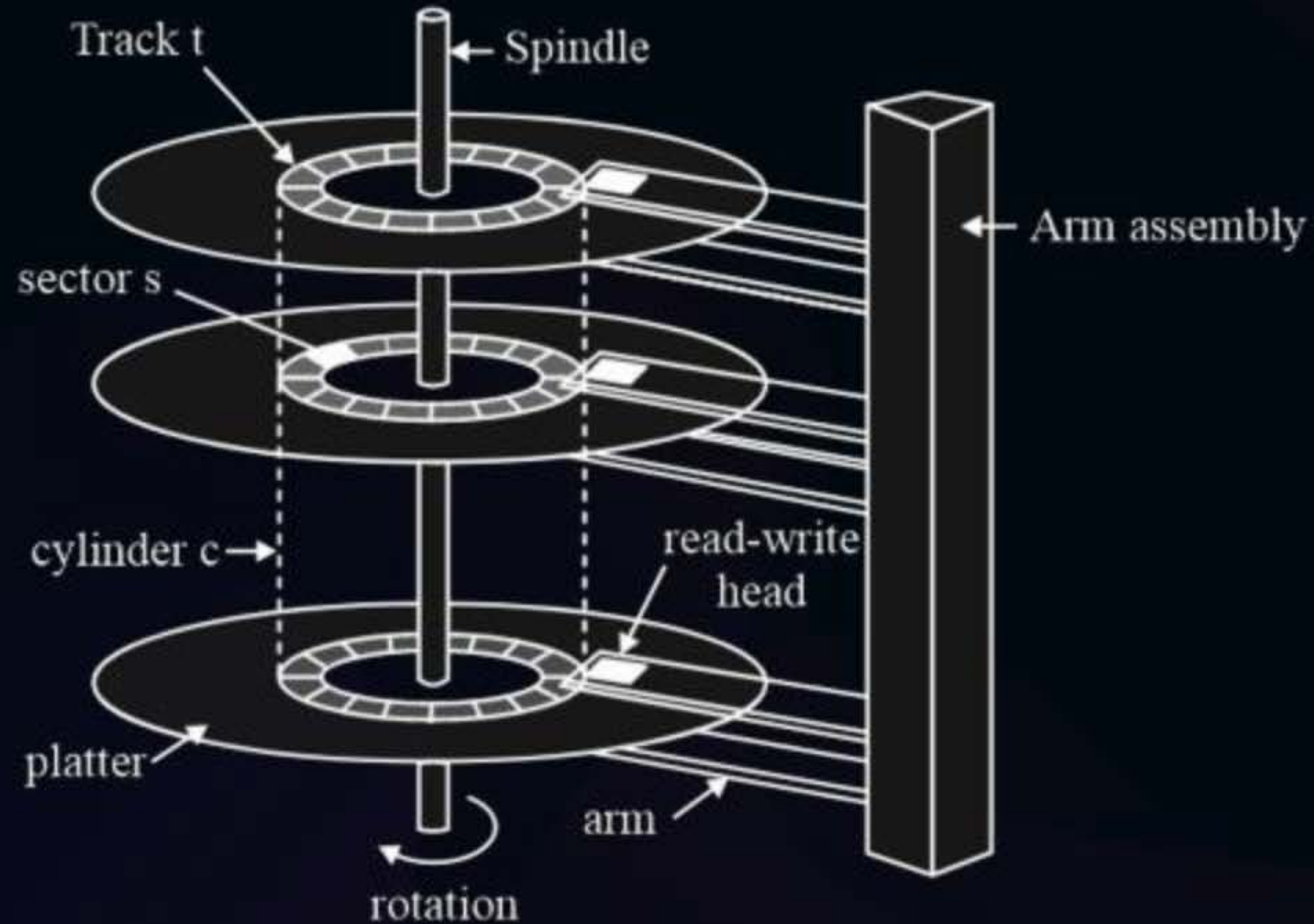


Topic : Cylinder



no. of cylinders in disk

= no. of tracks per surface



for sequential storage of file:-

sector transfer time $\Rightarrow$ for each sector

rotational latency $\Rightarrow$ once per track

seek time $\Rightarrow$ once per cylinder

ques no. of surfaces = 10

no. of sectors per track = 200

seek time = 10 ms

1 rotation time = 20 ms

1 file is stored on 4 cylinders sequentially in a cylinder.

$4 * 10 \Rightarrow \text{tracks} \Rightarrow 40 * 200 \text{ sectors}$

file transfer time =

solⁿ

$$= (4 * 10) + \left(40 * \frac{20}{2}\right) + \left(40 * \cancel{200} * \frac{20}{\cancel{200}}\right)$$

$$= 40 + 400 + 800 = 1240 \text{ ms}$$



2 mins Summary



Topic

Multiple Sector Access

Topic

Disk Cylinder

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Disk Addressing



Happy Learning

THANK - YOU

